

- 2 A McLaren Formula 1 car is fitted with a rocket engine that consumes fuel at a rate of 1.3 kg s^{-1} , producing a constant thrust that lasts for 15.0 s before the fuel runs out. The variation of the resultant force F acting on the car with the elapsed time t starting from the ignition of the rocket is shown in Fig. 2.1. The car is at rest at $t = 0$.

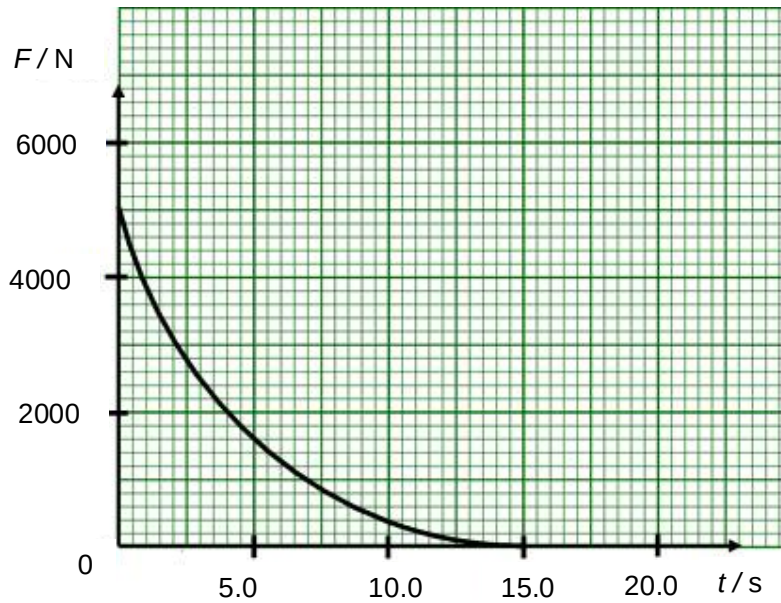


Fig. 2.1

- (a) Suggest why the resultant force acting on the car varies as shown in Fig. 2.1, even though the thrust is constant.

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..... [2]

(b) Calculate the speed at which the rocket exhaust gas leaves the rocket engine.

speed = m s^{-1} [3]

(c) Estimate the total impulse acting on the car.

impulse = kg m s^{-1} [3]

(d) The car has an initial total mass of 600 kg.

Calculate its total mass at $t = 15.0$ s.

mass = kg [1]

3 (a) State the first law of thermodynamics